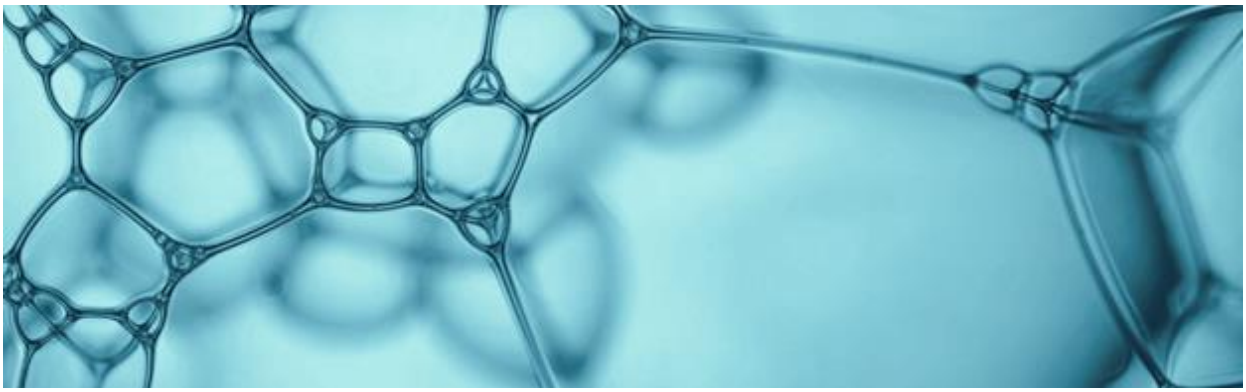

MACHINE LEARNING – 2

Business Project Report (Coded)

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PGP-DSBA

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Problem Statement - Coded Project

Business Context

Business communities in the United States are facing high demand for human resources, but one of the constant challenges is identifying and attracting the right talent, which is perhaps the most important element in remaining competitive. Companies in the United States look for hard-working, talented, and qualified individuals both locally as well as abroad.

The Immigration and Nationality Act (INA) of the US permits foreign workers to come to the United States to work on either a temporary or permanent basis. The act also protects US workers against adverse impacts on their wages or working conditions by ensuring US employers' compliance with statutory requirements when they hire foreign workers to fill workforce shortages. The immigration programs are administered by the Office of Foreign Labor Certification (OFLC).

OFLC processes job certification applications for employers seeking to bring foreign workers into the United States and grants certifications in those cases where employers can demonstrate that there are not sufficient US workers available to perform the work at wages that meet or exceed the wage paid for the occupation in the area of intended employment.

Objective

In FY 2016, the OFLC processed 775,979 employer applications for 1,699,957 positions for temporary and permanent labor certifications. This was a nine percent increase in the overall number of processed applications from the previous year. The process of reviewing every case is becoming a tedious task as the number of applicants is increasing every year.

The increasing number of applicants every year calls for a Machine Learning based solution that can help in shortlisting the candidates having higher chances of VISA approval. OFLC has hired the firm EasyVisa for data-driven solutions. You as a data scientist at EasyVisa have to analyze the data provided and, with the help of a classification model:

- Facilitate the process of visa approvals.
- Recommend a suitable profile for the applicants for whom the visa should be certified or denied based on the drivers that significantly influence the case status.

Data Description

The data contains the different attributes of the employee and the employer. The detailed data dictionary is given below.

Data Dictionary:

- **case_id:** ID of each visa application.
- **continent:** Information of continent the employee
- **education_of_employee:** Information of education of the employee
- **has_job_experience:** Does the employee has any job experience? Y= Yes; N = No
- **requires_job_training:** Does the employee require any job training? Y = Yes; N = No
- **no_of_employees:** Number of employees in the employer's company
- **yr_of_estab:** Year in which the employer's company was established
- **region_of_employment:** Information of foreign worker's intended region of employment in the US.
- **prevailing_wage:** Average wage paid to similarly employed workers in a specific occupation in the area of intended employment. The purpose of the prevailing wage is to ensure that the foreign worker is not underpaid compared to other workers offering the same or similar service in the same area of employment.
- **unit_of_wage:** Unit of prevailing wage. Values include Hourly, Weekly, Monthly, and Yearly.

- **full_time_position:** Is the position of work full-time? Y = Full-Time Position; N = Part-Time Position
- **case_status:** Flag indicating if the Visa was certified or denied

Executive Summary:

The increasing number of visa applications in the U.S. has made the approval process more complex and time-consuming. This analysis provides insights into key factors influencing visa certification outcomes, helping EasyVisa and the Office of Foreign Labor Certification (OFLC) streamline the application review process using data-driven strategies. The study identifies critical attributes of applicants, such as education level, job experience, job training requirements, and prevailing wage, that significantly affect visa approval chances.

Deliverables:

- Univariate and bivariate analyses of key variables influencing visa certification.
- Identification of patterns and insights from the data.
- Actionable business recommendations to improve the visa approval process.

Understanding the Data

Dataset Sample

Displaying the first 5 rows:

	case_id	continent	education_of_employee	has_job_experience	requires_job_training	no_of_employees	yr_of_estab	region_of_employment
0	EZYV01	Asia	High School	N	N	14513	2007	West
1	EZYV02	Asia	Master's	Y	N	2412	2002	Northeast
2	EZYV03	Asia	Bachelor's	N	Y	44444	2008	West
3	EZYV04	Asia	Bachelor's	N	N	98	1897	West
4	EZYV05	Africa	Master's	Y	N	1082	2005	South

Table 1: First 5 rows of the given dataset

Displaying the last 5 rows:

	case_id	continent	education_of_employee	has_job_experience	requires_job_training	no_of_employees	yr_of_estab	region_of_employment
25475	EZYV25476	Asia	Bachelor's	Y	Y	2601	2008	South
25476	EZYV25477	Asia	High School	Y	N	3274	2006	Northeast
25477	EZYV25478	Asia	Master's	Y	N	1121	1910	South
25478	EZYV25479	Asia	Master's	Y	Y	1918	1887	West
25479	EZYV25480	Asia	Bachelor's	Y	N	3195	1960	Midwest

Table 2: Last 5 rows of the given dataset

Structure and Types of Data

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 25480 entries, 0 to 25479
Data columns (total 12 columns):
 #   Column                 Non-Null Count  Dtype
---  ------                 --------------  -----
 0   case_id                25480 non-null  object
 1   continent              25480 non-null  object
 2   education_of_employee  25480 non-null  object
 3   has_job_experience     25480 non-null  object
 4   requires_job_training  25480 non-null  object
 5   no_of_employees        25480 non-null  int64
 6   yr_of_estab            25480 non-null  int64
 7   region_of_employment   25480 non-null  object
 8   prevailing_wage        25480 non-null  float64
 9   unit_of_wage           25480 non-null  object
 10  full_time_position     25480 non-null  object
 11  case_status            25480 non-null  object
dtypes: float64(1), int64(2), object(9)
memory usage: 2.3+ MB
```

Table 3: Checking the structure and type of data

OBSERVATIONS:

- There are **25480 rows** and **12 Columns** are present in the given datasets.
- It can be observed that no columns have less entries (less than 25480 rows) which indicates that there are **no missing values** in the given dataset.
- There are attributes of different types (int, float, object) in the data.
- There are **3 numerical columns** in the data and **9 categorical columns**.
- **Dependent variable** is the **case_status** which is of categorical type.

Statistical summary of the Data

Numerical Data

	count	mean	std	min	25%	50%	75%	max
no_of_employees	25480.000	5667.043	22877.929	-26.000	1022.000	2109.000	3504.000	602069.000
yr_of_estab	25480.000	1979.410	42.367	1800.000	1976.000	1997.000	2005.000	2016.000
prevailing_wage	25480.000	74455.815	52815.942	2.137	34015.480	70308.210	107735.513	319210.270

Table 4: Statistical summary of the numerical data

Observations:

Employee Count:

- The average number of employees is **5,667**, but there is significant variation with a standard deviation of **22,878**, suggesting some cases have many more employees than others.
- The median number of employees is 2,109, indicating that half of the companies have fewer than this number of employees, showing a skew towards larger firms.
- The range of employees is vast, with some companies reporting as few as **-26** employees (which might indicate data issues or missing data corrections), while others have as many as **602,069**.

Year of Establishment:

- The average establishment year is **1979**, which means many companies have been in business for quite a while.
- The oldest company in the dataset dates back to **1800**, while the most recent was founded in 2016.

Prevailing Wage:

- The average wage is around **$74,456**, but wages vary widely, as shown by the large standard deviation of **$52,816**.
- The median wage is about **$70,308**, meaning half of the businesses offer wages below this value and the other half above.
- The minimum wage reported is as low as **$2.13**, likely **an outlier or a special case**, and the highest wage reaches over **$319,210**.

Categorical Data

	count	unique	top	freq
case_id	25480	25480	EZYV01	1
continent	25480	6	Asia	16861
education_of_employee	25480	4	Bachelor's	10234
has_job_experience	25480	2	Y	14802
requires_job_training	25480	2	N	22525
region_of_employment	25480	5	Northeast	7195
unit_of_wage	25480	4	Year	22962
full_time_position	25480	2	Y	22773
case_status	25480	2	Certified	17018

Table 5: Statistical summary of the categorical data

```
case_id: ['EZYV01' 'EZYV02' 'EZYV03' ... 'EZYV25478' 'EZYV25479' 'EZYV25480']
continent: ['Asia' 'Africa' 'North America' 'Europe' 'South America' 'Oceania']
education_of_employee: ['High School' "Master's" "Bachelor's" 'Doctorate']
has_job_experience: ['N' 'Y']
requires_job_training: ['N' 'Y']
region_of_employment: ['West' 'Northeast' 'South' 'Midwest' 'Island']
unit_of_wage: ['Hour' 'Year' 'Week' 'Month']
full_time_position: ['Y' 'N']
case_status: ['Denied' 'Certified']
```

Table 6: Unique values of the categorical attribute

Observations:

- There are **25,480 unique cases** in this dataset, and each case has a unique identifier, with the most frequent identifier being "EZYV01," though it only appears once, confirming the uniqueness of each case. This attribute can be drop as it will not add value to the analysis.
- The dataset covers **6 continents**, with the majority of cases (over **16,861**) involving employees from **Asia**, indicating a strong representation of the Asian labor market in the data.
- The education level of employees is distributed across **4 categories**, with **Bachelor's degree** being the most common qualification, held by **10,234** employees, reflecting the high demand for college-educated workers.
- **14,802** employees are marked as having prior job experience, which constitutes more than half of the dataset, showing that the majority of positions may require experienced workers.

- The majority of cases (approximately **22,525**) do **not require job training**, which could suggest that these positions either expect employees to be fully qualified or involve minimal training needs.
- The data is divided into **5 regions**, with the **Northeast** being the most common region of employment for **7,195** employees, likely due to a high concentration of businesses and industries in that area.
- **Yearly wages** are the most commonly used wage unit, applied in **22,962** cases. This implies that long-term, stable employment is prevalent across the dataset.
- Majority of cases (**22,773**) are for **full-time positions**, highlighting that most of the roles offered involve full-time commitments.
- Of the **two case statuses**, **"Certified"** is the most frequent, with **17,018** cases approved, indicating a large number of applications have successfully met the necessary criteria.

Checking for missing values

case_id	0
continent	0
education_of_employee	0
has_job_experience	0
requires_job_training	0
no_of_employees	0
yr_of_estab	0
region_of_employment	0
prevailing_wage	0
unit_of_wage	0
full_time_position	0
case_status	0

Table 7: No. of Missing Values in each column

INFERENCE:

- There is no missing values.
- There is no duplicate values.
- There is no null values.

Exploratory Data Analysis

Univariate Analysis

Observation on no_of_employees:

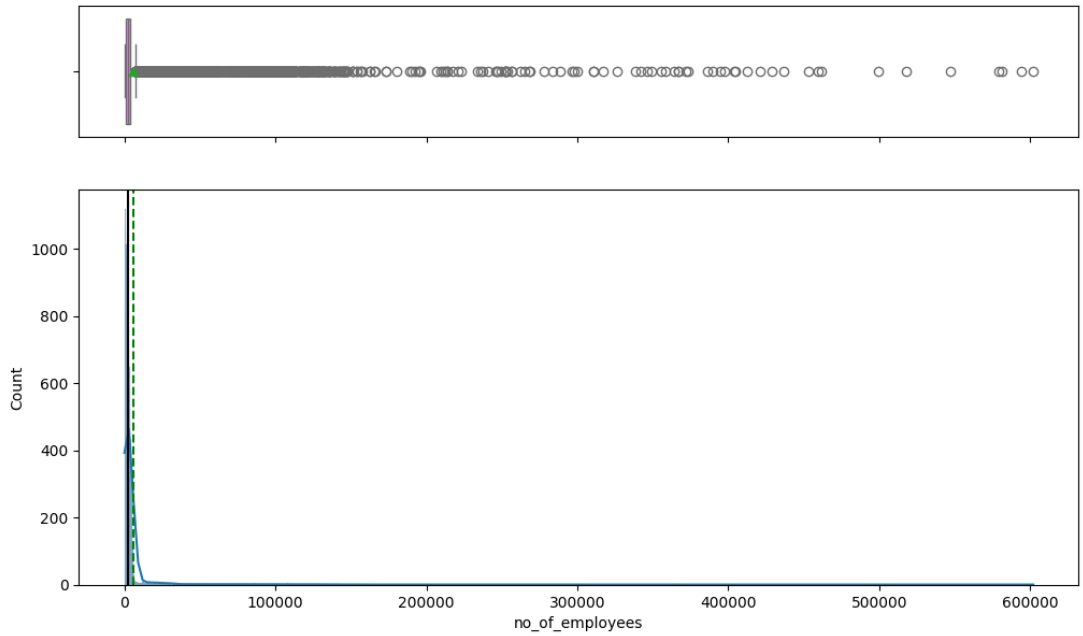

Fig 1: Histogram - boxplot of no_of_employees

OBSERVATIONS

- The **histogram** shows a **highly right-skewed distribution** for the number of employees, with most of the employers company employing a small number of workers.
- The dashed line on the histogram likely represents the **mean of the number of employees**, positioned far to the right of the peak, which suggests that a significantly raise the average number of employees.
- The **boxplot** reveals a significant number of **outliers on the right-hand side**, indicating that while most companies have a relatively low number of employees, some businesses are extremely large, skewing the data.
- Both plots show a high concentration of data near **zero employees**, which corresponds to the large number of smaller companies or possibly data entries with missing or incorrect values (especially since some values are negative, as observed in the earlier table).

Observation on yr_of_estab:

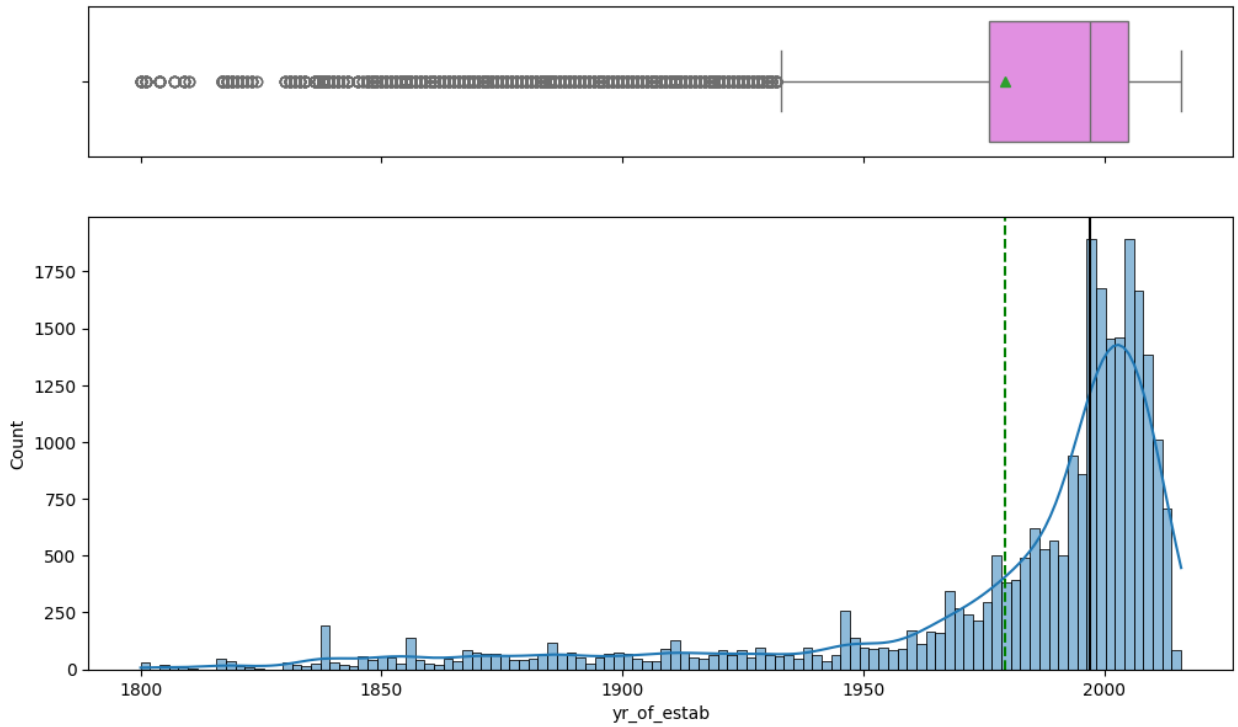

Fig 2: Histogram - boxplot of yr_of_estab

Observations

- The **Histogram** distribution has a **right-skewed** pattern, indicating that there are more recently founded companies compared to older ones.
- There are fewer companies established in the early 1900s or 1800s, but there are some noticeable spikes around those periods, possibly reflecting older industries.
- The **box-plot** includes **several outliers (dots)**, particularly for companies established before the 1850s. These outliers indicate the presence of very old companies in the dataset, though they represent a small portion.
- The interquartile range (IQR) is concentrated around the later years, likely after 1950, with a median year of establishment close to 1997 (which aligns with previous summary statistics).

Observation on prevailing_wage:

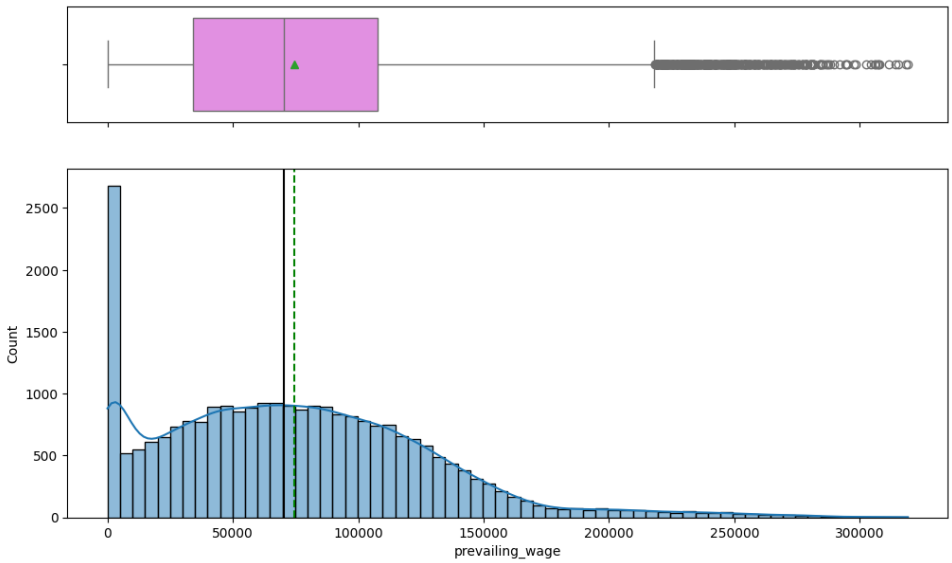

Fig 3: Histogram - boxplot of prevailing_wage

Observations

- The median wage, represented by the line within the box, is below the mean (indicated by the green triangle), suggesting that the data is right-skewed. This is further confirmed by the presence of outliers at the higher end of the wage spectrum.
- **There are many high-value outliers** (represented by dots), which shows that some companies offer significantly higher wages than the majority.
- The lower whisker is relatively short compared to the upper whisker, reflecting that the bulk of the wages are concentrated in the lower ranges.
- The histogram reveals that the prevailing wage distribution is right-skewed, with a large number of observations concentrated around the lower wage levels.
- There is a very high concentration of wages close to $0, indicating that a significant portion of the data includes lower-paying or part-time roles, or potentially incorrect entries.

Observation on continent:

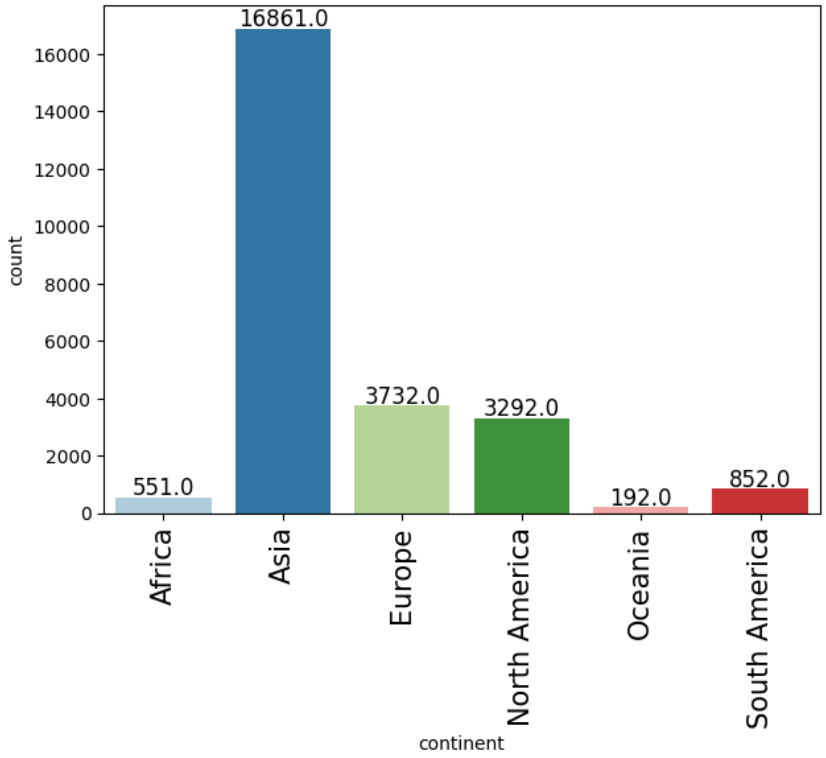

Fig 4: Barplot of continent

Observations

- **Asia** has the highest count, standing at **16,861**, which is significantly higher than any other continent.
- **Europe** follows with a count of **3,732**, making it the second-largest contributor.
- **North America** comes close to Europe with a count of **3,292**.
- **South America** has a relatively lower count of **852**.
- **Africa** has a count of **551**, reflecting a smaller contribution compared to other continents.
- **Oceania** has the lowest count among the continents, with only **192**.

Observation on education_of_employee:

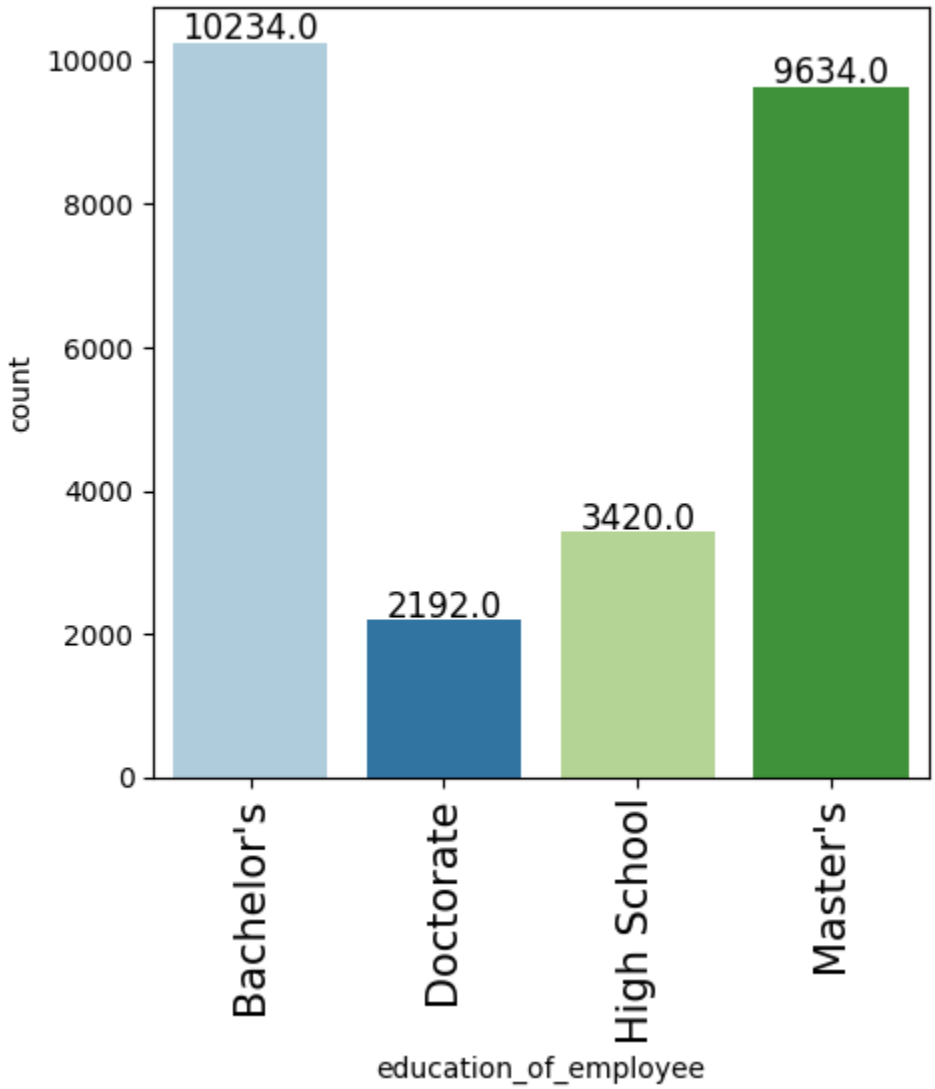

Fig 5: Barplot of education_of_employee

Observations

- **Bachelor's degree** holders represent the largest group, with a count of **10,234**, indicating that most employees have completed undergraduate education.

- Employees with a **Master's degree** are the second-largest group, with **9,634** individuals, which is only slightly lower than those with a Bachelor's.
- **High School** graduates account for **3,420** employees, a smaller but notable portion of the workforce.
- **Doctorate** holders are the smallest group, with a count of **2,192**, showing that fewer employees have pursued the highest level of academic achievement.

Observation on has_job_experience

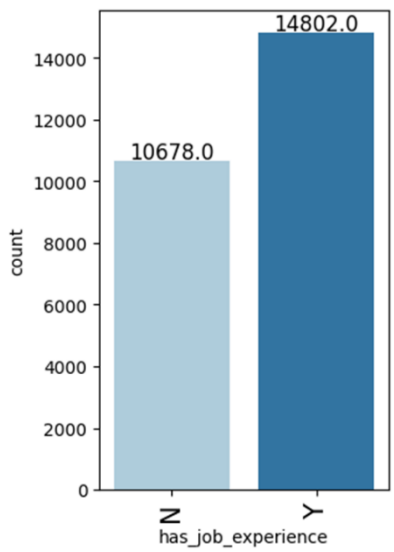

Fig 6: Barplot of has_job_experience

Observations

- A larger number of employees, **14,802**, have job experience, which is significantly higher compared to those without experience.
- Employees without job experience account for **10,678**, which, while substantial, is still lower than the group with experience.
- The difference between the two categories suggests that prior job experience is common among the majority of employees.

Observation on requires_job_training

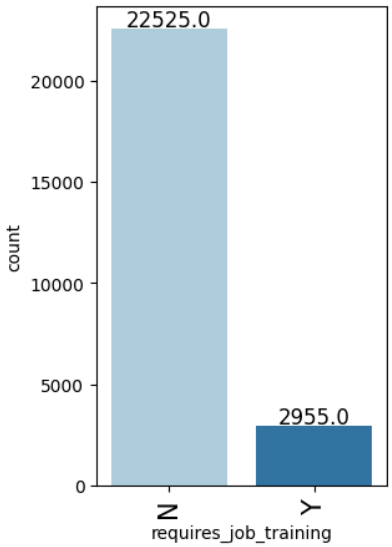

Fig 7: Barplot of requires_job_training

Observations

- The majority of the data, represented by a count of **22,525**, indicates that most positions do **not require job training**.
- Only a smaller segment, with a count of **2,955**, corresponds to positions that **do require job training**.
- This shows that a significant portion of the workforce in the dataset is hired for roles that don't need prior specialized training.

Observation on region_of_employment

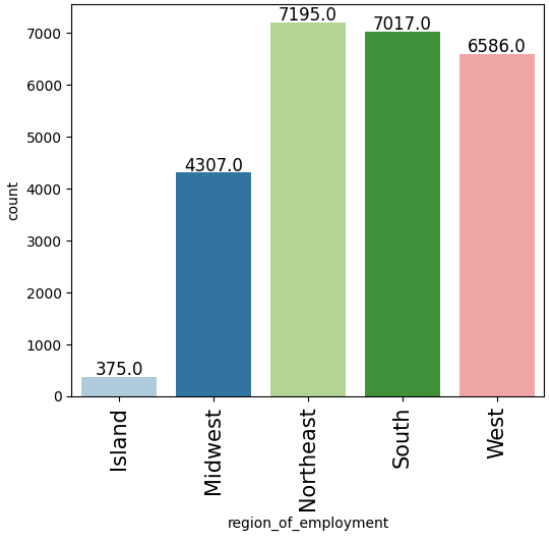

Fig 8: Barplot of region_of_employment

Observations

- **Northeast** and **South** are the leading regions for employment, with **7,195** and **7,017** individuals employed, respectively.
- The **West** also has a considerable count at **6,586**.
- The **Midwest** comes next with **4,307**, while the **Island** region shows the lowest employment figure, at only **375**.
- This suggests that employment opportunities are more concentrated in the Northeast, South, and West regions compared to the Midwest and Island regions.

Obsevarton on unit_of_wage

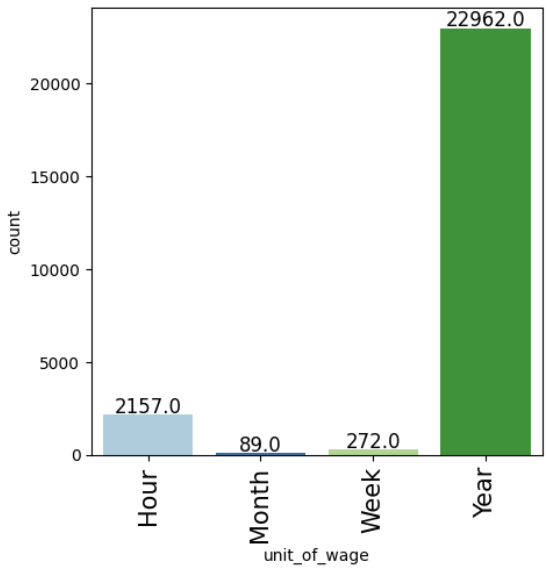

Fig 9: Barplot of unit_of_wage

Observations

- The majority of data points (22,962) are associated with the **Year** unit of wage. This suggests that most employees receive their wages on an annual basis.
- A significant number of employees (2,157) are paid on an hourly basis. This is the second most common wage unit after the annual salary.
- Very few employees are paid on a weekly (272) or monthly (89) basis compared to the other two categories.
- There is a large gap between yearly wages and other types, which highlights a tendency for most employees in this dataset to be compensated on a yearly basis.

Observation on full_time_position

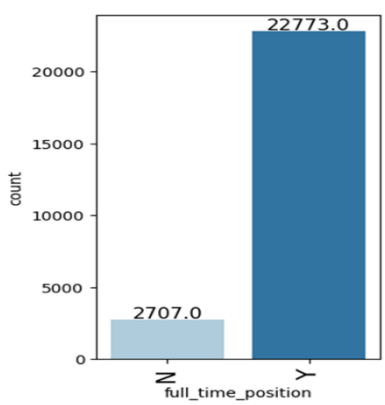

Fig 10: Barplot of full_time_position

Observations

- A large number of positions (**22,773**) are for **full-time roles**, while a smaller number (**2,707**) are for **non-full-time positions**.
- This reflects a strong preference or demand for full-time employment within this dataset.

Observation on case_status

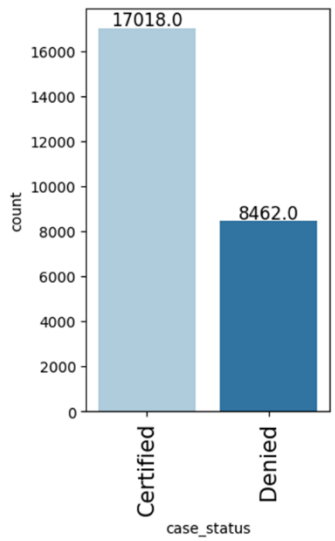

Fig 11: Barplot of case_status

Observations

- The bar chart represents the distribution of case statuses for two categories—Certified and Denied.
- There are significantly more certified cases (17,018) compared to denied cases (8,462), indicating a higher success rate for certifications.
- The number of certified cases is almost double that of denied cases, highlighting a trend toward higher certification approvals.

Bivariate Analysis

Correlation check

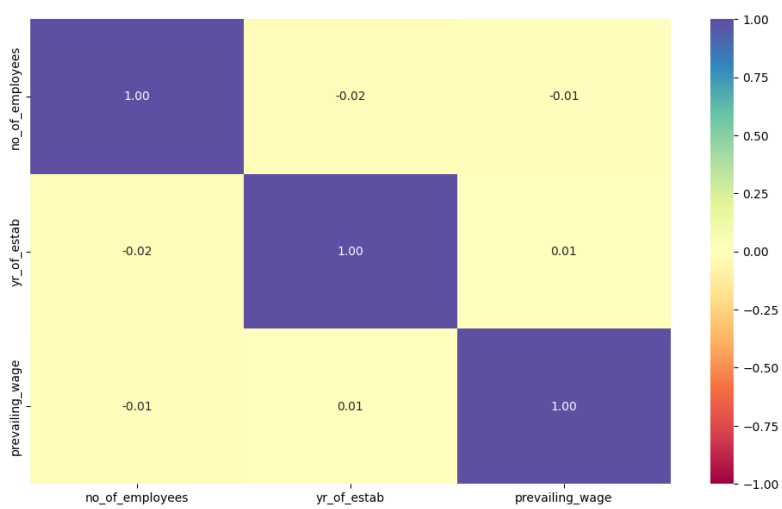

Fig 12: Heatmap of Numerical Variables

Observations

- The matrix indicates very weak correlations between the variables. The highest correlation is between the number of employees and itself, which is expected (1.00), while other correlations are near zero.
- The relationship between the number of employees and year of establishment is slightly negative (-0.02), meaning that older companies don't necessarily have more employees.
- Prevailing wage also has a weak correlation with the number of employees (-0.01) and year of establishment (0.01), suggesting that these factors do not significantly influence wage levels.

Observation on continent vs case_status

case_status	Certified	Denied	All
continent			
All	17018	8462	25480
Asia	11012	5849	16861
North America	2037	1255	3292
Europe	2957	775	3732
South America	493	359	852
Africa	397	154	551
Oceania	122	70	192

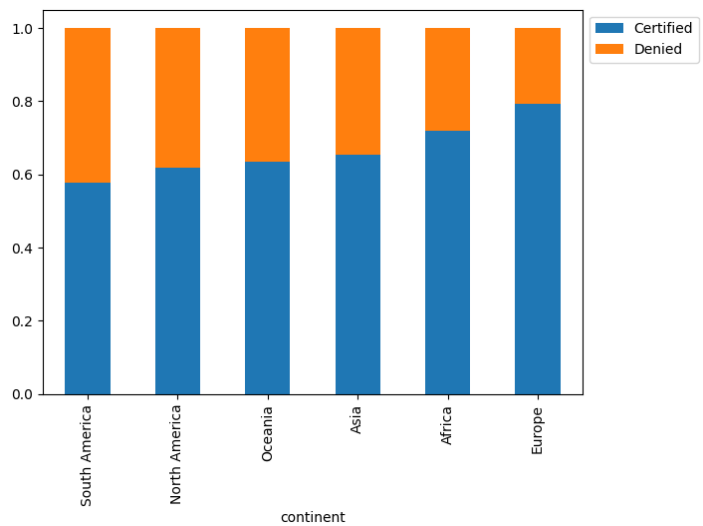

Fig 13: Barplot of continent vs case_status with count table

Observations

- A total of 25,480 cases are distributed across various continents, with Asia having the highest number of cases at 16,861, followed by Europe with 3,292. Oceania has the fewest cases, with 192 cases.

- The proportion of certified cases varies by continent. Africa has a relatively high certification rate, with most of its cases certified. In contrast, South America has a higher proportion of denied cases compared to other continents.
- Although Asia has the most cases, it also has a considerable number of denied cases, totaling 5,849, which is second only to its certified cases.
- Both Oceania and Africa have similar patterns in terms of certification and denial, with a higher proportion of certified cases compared to denied ones.
- The bar plot reveals that all continents, except South America, tend to have a greater number of certified cases than denied cases, indicating a generally positive outcome across regions.

Observation on education_of_employee vs case_status

case_status	Certified	Denied	All
education_of_employee			
All	17018	8462	25480
Bachelor's	6367	3867	10234
High School	1164	2256	3420
Master's	7575	2059	9634
Doctorate	1912	280	2192

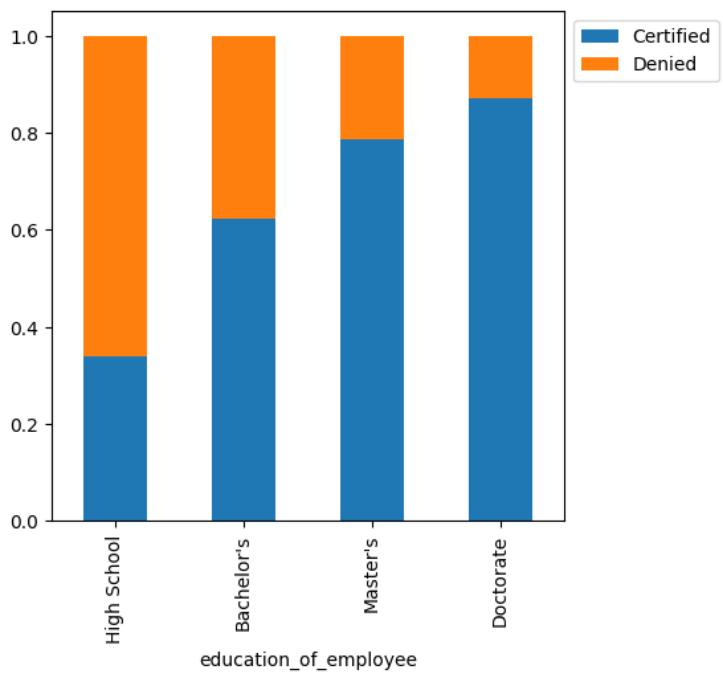

Fig 14: Barplot of education_of_employee vs case_status with count table

Observations

- Across all education levels, the majority of cases are certified. The certification rate is particularly high for individuals with a Doctorate, where certified cases make up a significant majority.
- Among the educational levels, Bachelor's degree holders experience a relatively higher rate of denials, with 3,867 denied cases out of a total of 10,234. This is a notable difference compared to other education levels.
- High School graduates have the lowest number of certified cases (1,164), and a substantial portion of their cases are denied (2,256), indicating a challenging outcome for this group.
- Master's and Doctorate degree holders show consistent patterns, with high certification rates and low denial rates. This suggests that higher education levels tend to correlate with more favorable outcomes.
- The bar plot visually confirms that higher educational attainment correlates with higher certification rates. Individuals with advanced degrees (Master's and Doctorate) have noticeably better outcomes compared to those with lower levels of education.

Observation on has_job_experience vs case_status

case_status	Certified	Denied	All
has_job_experience			
All	17018	8462	25480
N	5994	4684	10678
Y	11024	3778	14802

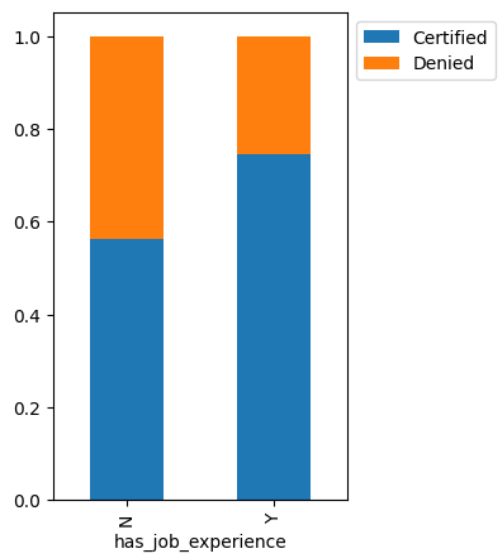

Fig 15: Barplot of has_job_experience vs case_status with count table

Observations

- Applicants with **job experience (Y)** have a higher certification rate compared to those without job experience. Out of 14,802 applicants with job experience, 11,024 were certified, while only 3,778 were denied. In contrast, out of 10,678 applicants without job experience, 5,994 were certified, and 4,684 were denied.
- Having job experience positively impacts visa approval chances, with experienced candidates being more likely to get certified.

Observation on requires_job_training vs case_status

case_status	Certified	Denied	All
requires_job_training			
All	17018	8462	25480
N	15012	7513	22525
Y	2006	949	2955

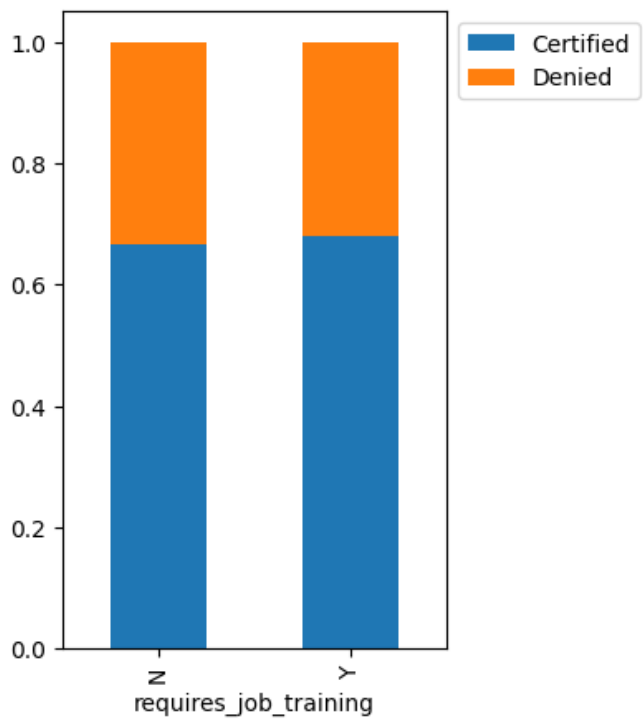

Fig 16: Barplot of requires_job_training vs case_status with count table

Observations

- The requirement of **job training** does not appear to have a substantial impact on certification outcomes. For applicants not requiring training (23,525 total), 15,012

were certified, and 7,513 were denied. For those requiring job training (2,955 total), 2,006 were certified, and 949 were denied.

- The requirement for job training has a minimal influence on visa approval decisions, with slightly more certifications for those not requiring training.

Observation on region_of_employment vs case_status

case_status	Certified	Denied	All
region_of_employment			
All	17018	8462	25480
Northeast	4526	2669	7195
West	4100	2486	6586
South	4913	2104	7017
Midwest	3253	1054	4307
Island	226	149	375

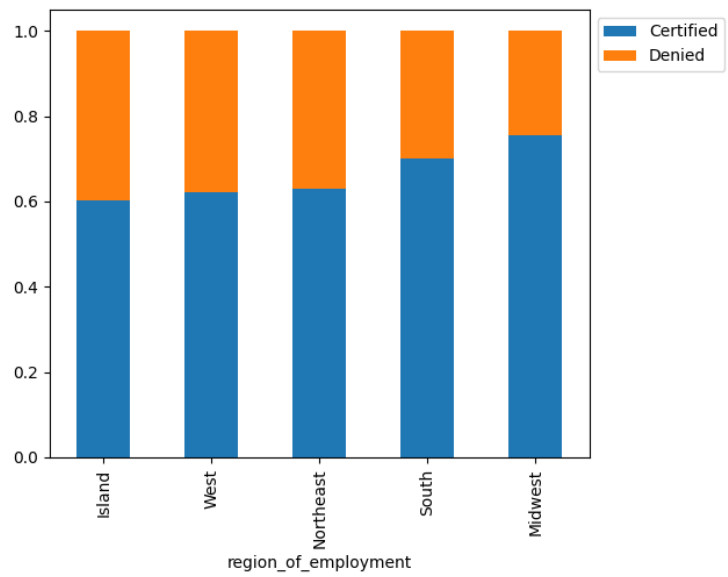

Fig 17: Barplot of region_of_employment vs case_status with count table

Observations

- The region of employment has a noticeable impact on visa certification outcomes. The **Midwest and West regions have higher certification rates** compared to other regions. For example, in the Midwest (43,807 total applicants), 32,253 were certified, and 10,654 were denied. Similarly, in the West (6,856 total), 4,410 were certified, and 2,446 were denied.

- The Island region has the lowest number of applications but also the lowest certification rate.

- Employers in regions like the Midwest and West tend to have higher certification rates, while regions such as the Island have lower success rates in visa approvals.

Observation on full_time_position vs case_status

case_status	Certified	Denied	All
full_time_position			
All	17018	8462	25480
Y	15163	7610	22773
N	1855	852	2707

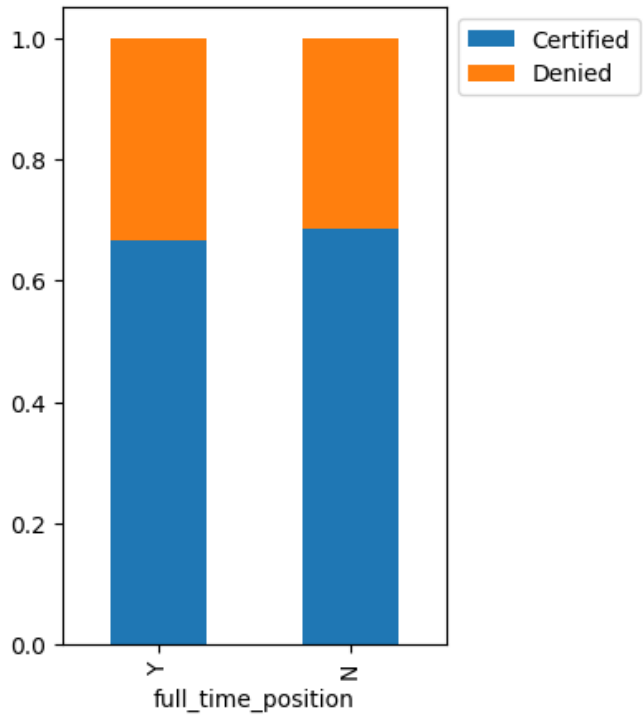

Fig 18: Barplot of full_time_position vs case_status with count table

Observations

- The proportion of certified cases is greater than denied cases for both full-time (Y) and part-time (N) positions. This is reflected by the blue sections of the stacked bars being larger in both cases. However, full-time positions have a slightly higher proportion of denials compared to part-time positions.

- There is a larger overall count of cases for full-time positions (Y) compared to part-time (N), as shown in the count table at the top of the figure (15,163 certified and 7,610 denied for full-time, versus 1,855 certified and 852 denied for part-time).

Observation on no_of_employees vs case_status

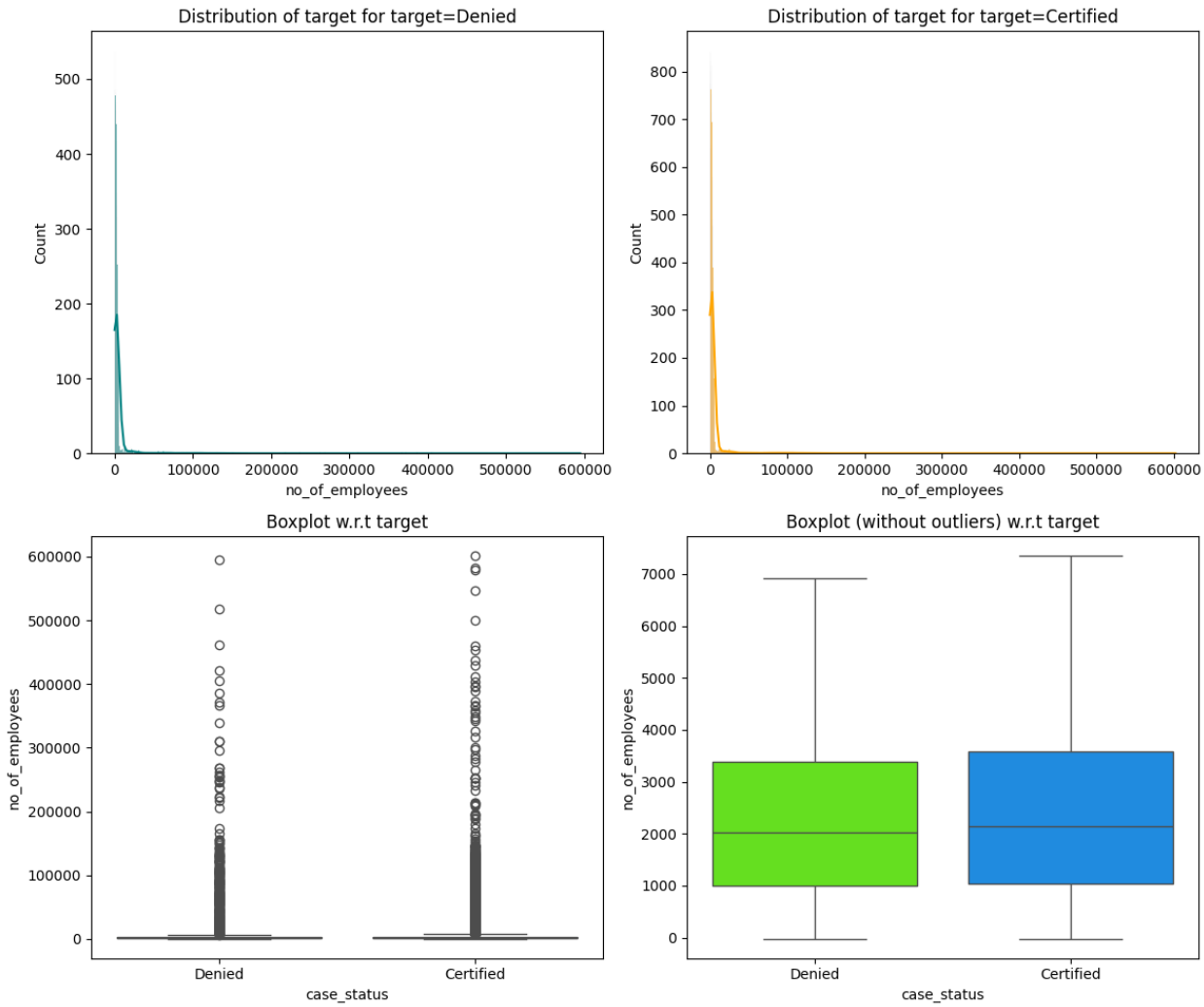

Fig 19: Histogram-boxplot of no_of_employees vs case_status

Observations

Distribution (Denied vs Certified):

- The distribution of the number of employees is highly skewed for both denied and certified cases, with most values concentrated below 100,000 employees.
- There are some extreme outliers with firms having over 500,000 employees, but these are rare.

Boxplot Observations:

- The median number of employees for both denied and certified cases seems quite close, with only slight differences.
- The spread of employees is very similar for both case statuses, but the presence of outliers in the upper ranges indicates variability in larger organizations.
- The size of the organization (measured by employees) does not appear to be a major differentiator between denied and certified applications, though there are more outliers in both cases.

Observation on yr_of_estab vs case_status

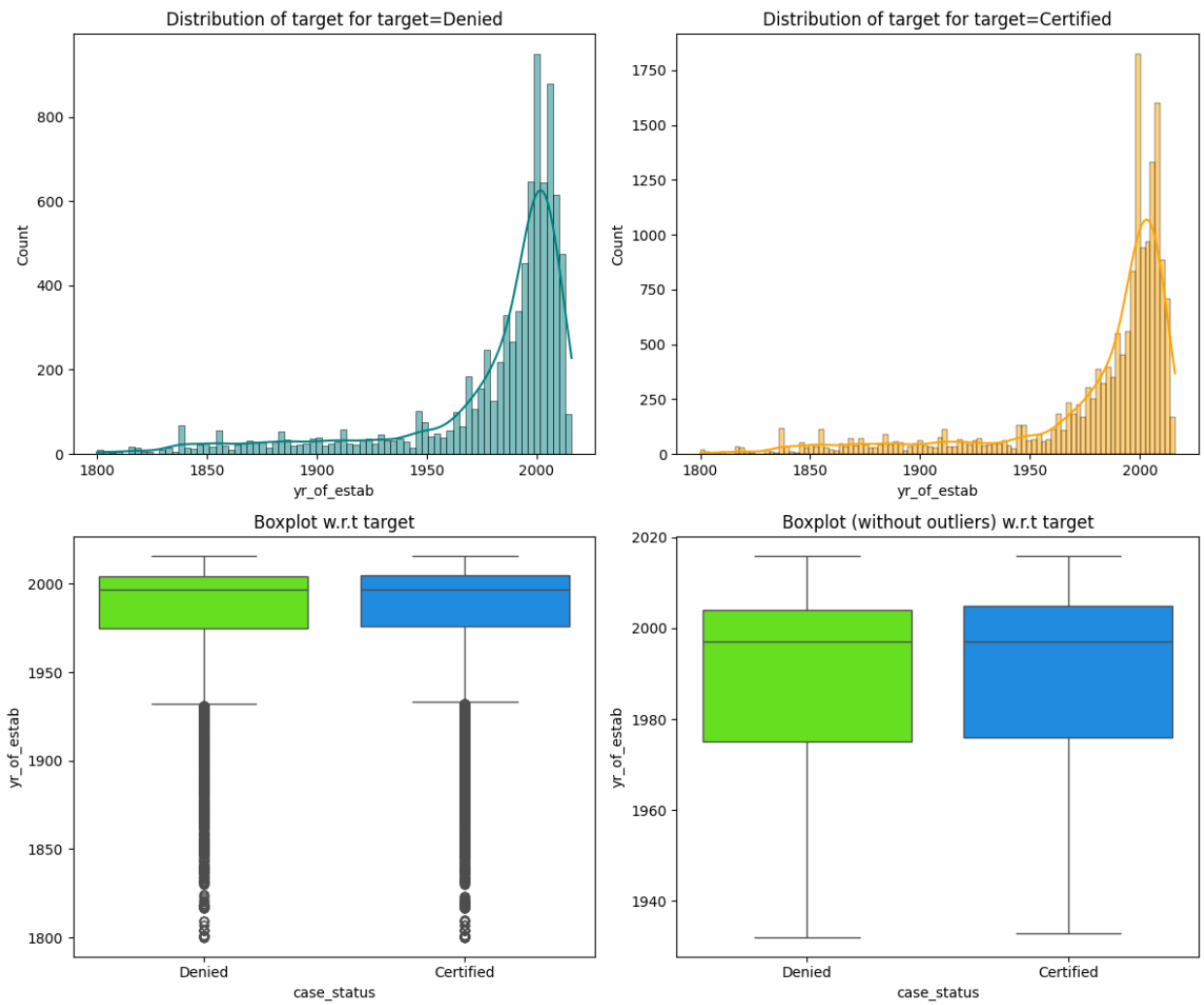

Fig 20: Histogram-boxplot of yr_of_estab vs case_status

Observations

Distribution (Denied vs Certified):

- Both denied and certified cases show a similar distribution trend where most firms were established between 1950 and 2000.
- Older establishments (from 1800 to 1900) have significantly fewer applications in both categories.

Boxplot Observations:

- The median year of establishment for both denied and certified cases is around the year 2000, indicating that younger firms dominate these applications.
- The spread is slightly wider for certified cases, but overall, companies founded in the 20th century make up most of the data.
- Younger companies (established post-1950) tend to dominate applications, and there is little difference between denied and certified cases based on the company's age.

Observation on prevailing_wage vs case_status

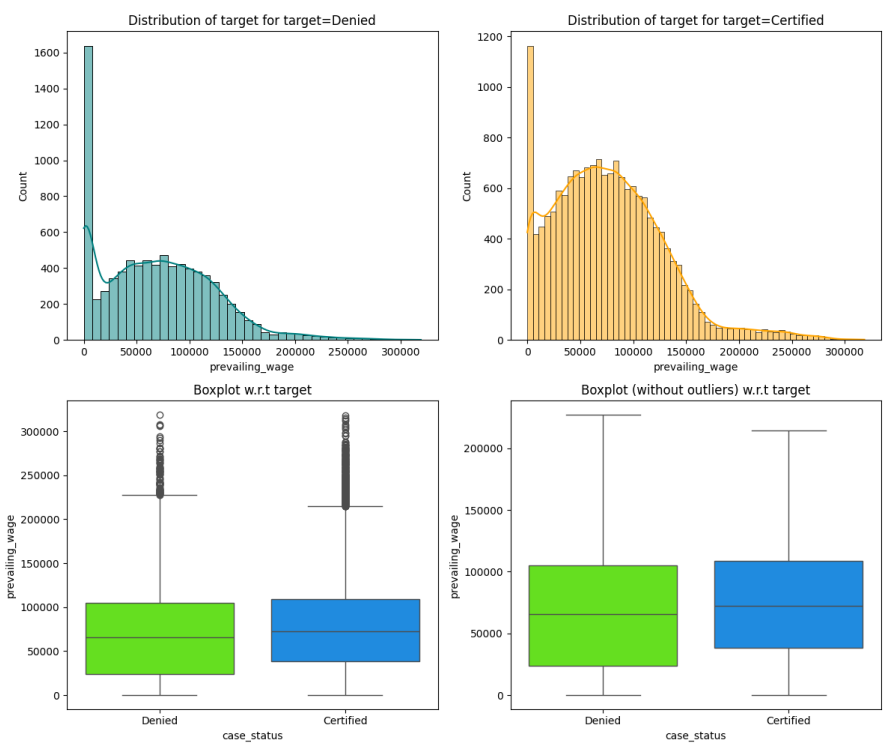

Fig 21: Histogram-boxplot of prevailing_wage vs case_status

Observations

Distribution (Denied vs Certified):

- The prevailing wage distributions for both denied and certified cases exhibit a skew toward lower wage ranges, with the majority of cases concentrated below $100,000.
- For certified cases, a slightly larger number of applications fall in higher wage ranges compared to denied cases.

Boxplot Observations:

- Both denied and certified cases have a similar median prevailing wage, around $60,000 to $70,000.
- There is a greater spread of prevailing wage for certified cases, suggesting that certified cases include a wider range of wage levels.
- Outliers in both cases show a few applications with very high prevailing wages, surpassing $200,000.
- A key differentiator may be the wage offered, where certified cases tend to encompass a broader range of salaries, including more cases in higher wage brackets.

EDA INSIGHTS

- The majority of visa applications are certified, with a significant portion denied. This highlights the necessity for more rigorous analysis to understand the drivers of denial.
- The data shows a wide range of education levels among applicants, with Bachelor's and Master's degrees being the most common. Understanding the influence of education on visa approval can be insightful.
- A smaller proportion of applicants require job training, indicating that the majority of positions seek candidates who do not need additional training.
- Applicants with higher education levels, such as Master's and Ph.D. degrees, have a higher likelihood of certification compared to those with lower education levels, like High School or Associate's degrees. This suggests that higher education is a significant factor in visa approvals.

- Applicants with job experience are more frequently certified than those without experience. This reinforces the importance of work experience in the approval process.

- Applicants who do not require job training have a higher certification rate, indicating that employers prefer candidates who are ready to start working without additional training.

- Certified cases tend to have a higher prevailing wage compared to denied cases. This suggests that offering competitive wages is associated with a higher likelihood of visa approval.

Data Preprocessing

Outlier Check

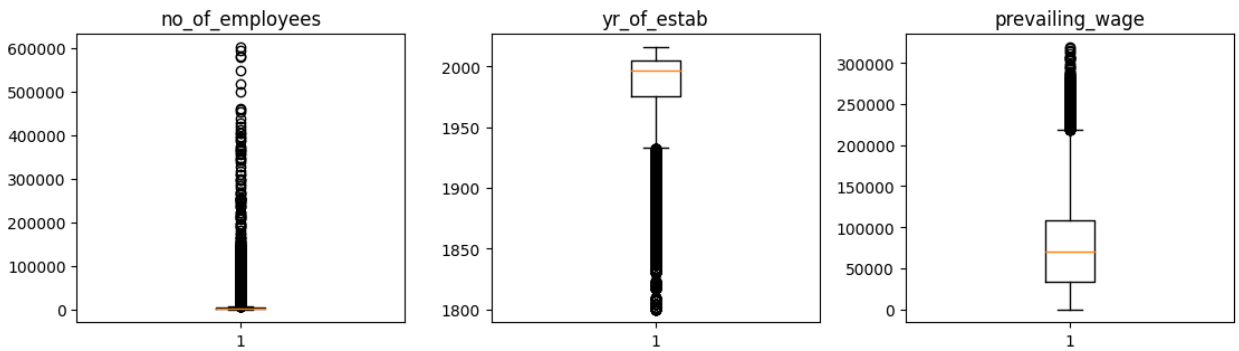

Fig 22: Barplot of all numerical variable

Observations

- **no_of_employees:** The boxplot reveals a significant number of outliers on the right-hand side, indicating that while most companies have a relatively low number of employees, some businesses are extremely large.

There is high concentration of data near zero employees, which corresponds to the large number of smaller companies or possibly incorrect data entries (especially since some values are negative, as observed in the earlier table). As we don't know the meaning of negative sign and the propotion is also in very small quantity so we can neglect the outliers.

- **yr_of_estab:** In this case, the outliers indicate the presence of very old companies in the dataset, though they represent a small portion are normal. And therefore, no need to outlier treatment in this case.

- **prevailing_wage:** It can be observed that there are many high-value outliers which shows that some companies which can be due to different unit of wages, i.e. Hourly/ Weekly/ Monthly/ Yearly. The outliers in this case are also normal. Therefore, no need to outlier treatment.

INFERENCE

- There are no outliers that require treatment.
- Case_Status is defined as dependent variable.
- As We are predicting the applicants for whom the visa should be certified or denied. And So, Case_Status" column converted from categorical to numerical (int) for building models with Denied as 0 & Certified as 1.
- The data is splitted into training and validation sets for further evaluation.-
    - The first train_test_split splits the data into training and temporary sets, with 20% of the data being set aside for testing.
    - The second train_test_split further divides the entire dataset into test and validation sets. However, it uses the entire dataset (features and target) to create the test and validation splits, which means that this split includes the data already assigned to the training set.

Model Building

Next, I will build Models using Bagging, Random Forest, Ada Boost, Gradient Boosting, XGBoost methods

- Let's build and check the model performance.

Model evaluation criterion

Model can make wrong predictions as:

- **False Positive(FP):** Predicting an applicant's visa is marked as "certified" but it should have been "denied."
- **False Negative(FN):** Predicting an applicant's visa is marked as "denied" but it should have been "certified."

To reduce the losses which might occur due false negative the recall need to be maximized, greater the recall score higher are the chances of minimizing the False Negatives.

To reduce the losses which might occur due false negative the recall need to be maximized**, greater the recall score higher are the chances of minimizing the False Negatives**.

- Model has been built.

  Below before and after sampling details mentioned below.

Over-sampled Data

```
Before Oversampling, counts of label 'Yes': 13614
Before Oversampling, counts of label 'No': 6770

After Oversampling, counts of label 'Yes': 13614
After Oversampling, counts of label 'No': 13614

After Oversampling, the shape of train_X: (27228, 21)
After Oversampling, the shape of train_y: (27228,)
```

Under-sampled Data

```
Before Under Sampling, counts of label 'Yes': 13614
Before Under Sampling, counts of label 'No': 6770

After Under Sampling, counts of label 'Yes': 6770
After Under Sampling, counts of label 'No': 6770

After Under Sampling, the shape of train_X: (13540, 21)
After Under Sampling, the shape of train_y: (13540,)
```

Model Performance - Original Data

```
Training Performance:

Bagging: 0.9847216101072426
Random forest: 1.0
AdaBoost: 0.8859997061848097
GradientBoosting: 0.876156897311591
XGBoost: 0.9305127075069781

Validation Performance:

Bagging: 0.9410936399906126
Random forest: 0.9683172964092936
AdaBoost: 0.8946256747242431
GradientBoosting: 0.883126026754283
XGBoost: 0.9185637174372213
```

Table 8: Original Model Performance based on recall score

Observations

- The **Random Forest** and **XGBoost** models seem to perform best overall, achieving high recall score both in training and validation phases.

- **AdaBoost** and **Gradient Boosting** maintain more moderate and consistent performance, with less variance between training and validation.
- **Bagging** also performs well but shows a slightly more noticeable drop between training and validation.

Model Performance - Oversampled Data -

```
Training Performance:

Bagging: 0.9847216101072426
Random forest: 1.0
AdaBoost: 0.8859997061848097
GradientBoosting: 0.876156897311591
XGBoost: 0.9305127075069781

Validation Performance:

Bagging: 0.9410936399906126
Random forest: 0.9683172964092936
AdaBoost: 0.8946256747242431
GradientBoosting: 0.883126026754283
XGBoost: 0.9185637174372213
```

Table9: Over-sampled Model Performance based on recall score

Observations

- **Random Forest** stands out as having the best performance in both training and validation datasets.
- **Bagging** and **XGBoost** also perform very well and generalize effectively, showing a good balance between training and validation performance.
- **AdaBoost** and **Gradient Boosting** exhibit slightly lower performances compared to the other models, with **Gradient Boosting** in particular showing a noticeable drop in validation accuracy.
- **Random Forest**, **Bagging**, and **XGBoost** seem to be the top models for this dataset, particularly in terms of generalization, while **AdaBoost** and **Gradient Boosting** may need further tuning to improve their performance.

Model Performance – Under-sampled Data

```
Training Performance:

Bagging: 0.9716395864106352
Random forest: 1.0
AdaBoost: 0.7276218611521418
GradientBoosting: 0.7515509601181684
XGBoost: 0.8462333825701625

Validation Performance:

Bagging: 0.7714151607603849
Random forest: 0.814362825627787
AdaBoost: 0.7204881483219902
GradientBoosting: 0.7470077446608777
XGBoost: 0.7580380192443088
```

Table 10: Under-sampled Model Performance based on recall score

Observations

- **Random Forest** once again outperforms the other models in both training and validation sets, showing the best generalization ability despite potential overfitting concerns from the perfect training score.
- **Bagging** also performs well, although its validation score shows a noticeable drop compared to training.
- **XGBoost** performs better in training compared to AdaBoost and Gradient Boosting, but its validation score is lower than expected, suggesting a gap in generalization.
- Both **AdaBoost** and **Gradient Boosting** exhibit weaker performance, with relatively close training and validation scores, indicating that they might not be able to fully capture the complexity of the dataset.

Model Performance Improvement

Hyperparameter Tuning

Tuning AdaBoost using original data

Train sets

	Accuracy	Recall	Precision	F1
0	0.737	0.914	0.748	0.823

Table 11: Tuned AdaBoost using original data model performance on training set

Test sets

	Accuracy	Recall	Precision	F1
0	0.739	0.916	0.749	0.824

Table 12: Tuned AdaBoost using original data model performance on test set

OBSERVATIONS

- The tuned AdaBoost model performs consistently well across both training and test sets, especially excelling in recall, which suggests it does a great job in minimizing false negatives.
- The balance between precision and recall, reflected in a high F1 score, indicates strong, reliable overall performance.

Tuning Ada Boost using under-sampled data

Train sets

	Accuracy	Recall	Precision	F1
0	0.739	0.916	0.749	0.824

Table 13: Tuned AdaBoost using original data model performance on training set

Test sets

	Accuracy	Recall	Precision	F1
0	0.677	0.654	0.827	0.731

Table 14: Tuned AdaBoost using original data model performance on test set

Observations

- While the model shows good performance on the training set with balanced precision and recall, the **test set performance declines** notably, especially in recall.
- The high precision indicates that the model is good at avoiding false positives, but the lower recall suggests it is struggling to identify a substantial number of true positives.
- This drop in performance may be a consequence of using **under-sampled data**, which can limit the model's ability to generalize effectively.

Tuning Gradient Boosting using under-sampled data

Train sets

	Accuracy	Recall	Precision	F1
0	0.715	0.767	0.695	0.729

Table 15: Tuned Gradient Boosting using under-sampled data model performance on training set

Test sets

	Accuracy	Recall	Precision	F1
0	0.728	0.764	0.818	0.790

Table 16: Tuned Gradient Boosting using under-sampled model performance on test set

Observations

- The model shows solid performance with a notable improvement in precision on the test set. This suggests that tuning Gradient Boosting on under-sampled data helped in reducing bias while maintaining a good recall and balance between other metrics.

Tuning Gradient Boosting using original data

Train sets

	Accuracy	Recall	Precision	F1
0	0.678	0.995	0.676	0.805

Table 17: Tuned Gradient Boosting using original data model performance on training set

Test sets

	Accuracy	Recall	Precision	F1
0	0.679	0.995	0.677	0.806

Table 18: Tuned Gradient Boosting using original data model performance on test set

Observations

- The model exhibits a strong tendency toward recall, performing well in identifying positive cases but at the cost of more false positives, as reflected by its moderate precision.
- The consistent accuracy, precision, and F1 score across the training and test sets show that the model generalizes well but could benefit from improvements in precision if reducing false positives is a priority.

Tuning XGBoost Model with Original data

Train sets

	Accuracy	Recall	Precision	F1
0	0.668	1.000	0.668	0.801

Table 19: Tuned XGBoost Model with Original data model performance on training set

Validation sets

	Accuracy	Recall	Precision	F1
0	0.669	1.000	0.669	0.802

Table 20: Tuned XGBoost Model with Original data model performance on test set

Observations

- The XGBoost model displays excellent recall but moderate precision, which indicates a high sensitivity toward positive cases at the cost of false positives. The balanced F1 score reflects this trade-off.

- The consistency in accuracy, recall, precision, and F1 across both training and test sets highlights that the model is generalizing well but may need further tuning to improve precision if reducing false positives is crucial.

Model Performance Comparison and Final

Model Selection

Training Model Performance comparison

Training performance comparison:

	Gradient boosting trained with Undersampled data	Gradient boosting trained with Original data	AdaBoost trained with Undersampled data
Accuracy	0.715	0.678	0.694
Recall	0.767	0.995	0.657
Precision	0.695	0.676	0.709
F1	0.729	0.805	0.682

Table 21: Training Model Performance comparison table

Validation Model comparison

Validation performance comparison:

	Gradient boosting trained with Undersampled data	Gradient boosting trained with Original data	AdaBoost trained with Undersampled data
Accuracy	0.728	0.679	0.677
Recall	0.764	0.995	0.654
Precision	0.818	0.677	0.827
F1	0.790	0.806	0.731

Table 22: Validation Model Performance comparison table

Observations

- **Gradient Boosting (Undersampled data)** performs well in terms of recall, which may be desirable if identifying true positives is critical. Its strong generalization performance on the test data with a high F1-score indicates it is handling the undersampled data well.
- **Gradient Boosting (Original data)** is excellent at recall, identifying most of the positive instances, but its lower precision indicates a high number of false positives, which may not be suitable if precision is critical.
- **AdaBoost (Undersampled data)** strikes a better balance with higher precision across both training and test sets, making it a good candidate when reducing false positives is essential.

Final model selection

- If **recall** (identifying as many true positives as possible) is the priority, **Gradient Boosting (Original data)** may be the preferred choice.
- If **precision** (reducing false positives) is more critical, **AdaBoost (Undersampled data)** would be the better option.
- The **Gradient Boosting (Undersampled data)** seems to be the best model overall offers a balanced performance with strong generalization, making it an all-around solid choice for many cases.

Selected Model Feature Performance

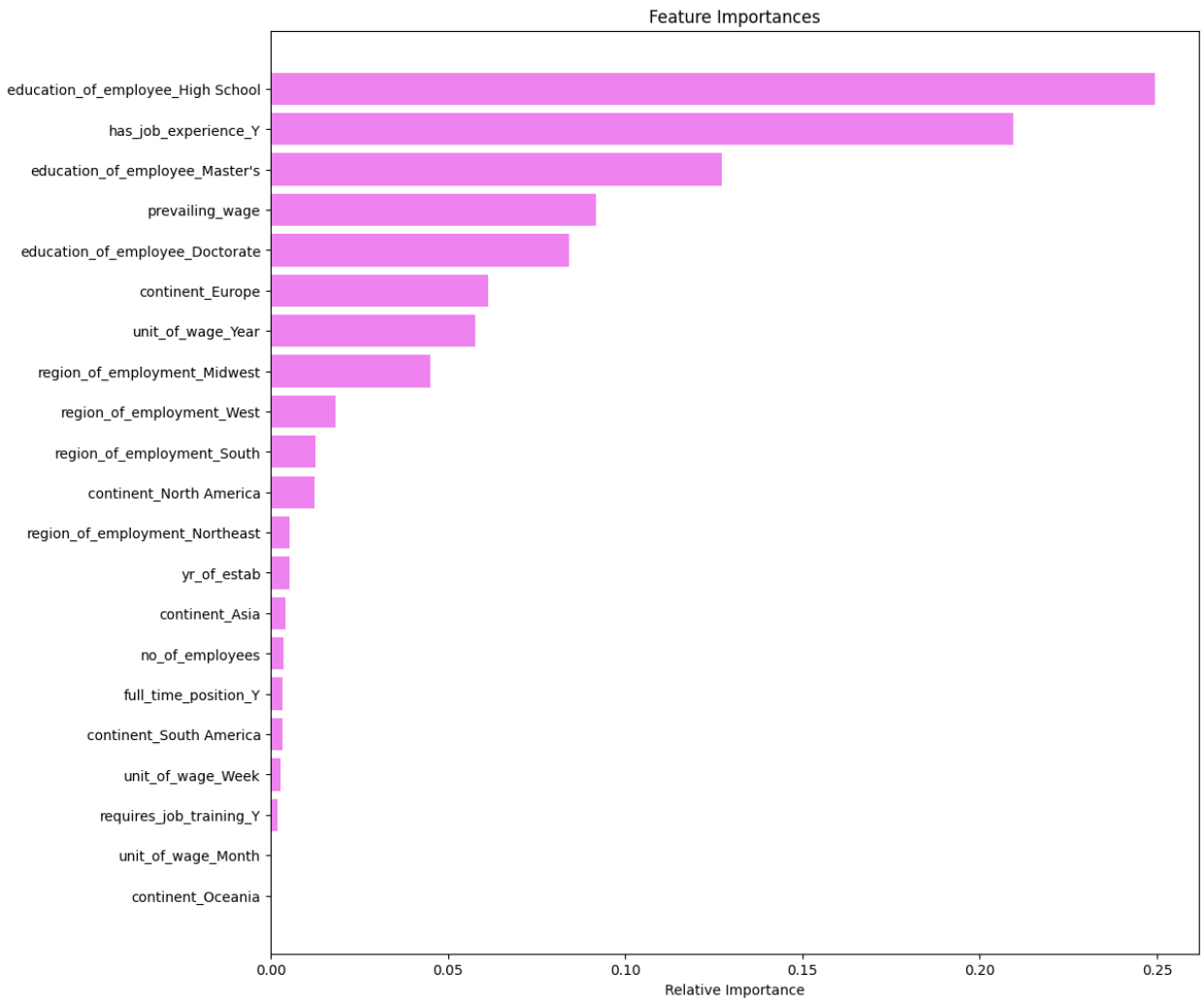

Fig 23: Feature importance

Observations

- The most important feature in this model is education level (High School), which has the highest relative importance, suggesting that employees with a high school education are significantly influencing the target outcome.

- Job experience also plays a crucial role, indicated by the second-highest importance. It implies that candidates with prior job experience are highly relevant to the model's predictions.

- Apart from high school education, Master's and Doctorate education levels are also among the top features, indicating that the model assigns a lot of weight to the educational qualifications of employees.

- The prevailing wage is another key factor, ranked fourth, suggesting that wage levels have a strong influence on the outcome (likely related to the certification decision).

- Continent (Europe) and regions of employment (Midwest and West) are relatively important, showing that the location where the employee is based has a moderate effect on the model's predictions.

- Features like unit of wage (Year) and year of establishment also have some importance, though much lower than educational and job-related factors.

- The model places the highest weight on education, job experience, and wage levels, with geographic location playing a secondary but still relevant role.

Business Recommendations and Insights

- **Education**, particularly at the high school, Master's, and Doctorate levels, plays a critical role in visa approval decisions. Companies should prioritize applicants with higher educational qualifications as they have a higher likelihood of visa certification.
- EasyVisa's model suggests that educational background is one of the top determinants of visa outcomes. Hence, companies can streamline their recruitment by focusing on highly educated candidates when applying for foreign labor certifications.
- **Job experience** is the second most influential factor in the visa certification process. Candidates with previous job experience are more likely to have their visa applications approved. Employers should seek candidates who have demonstrable job experience, as they are more likely to meet the approval criteria. Additionally, the OFLC should encourage companies to prioritize skilled workers with experience to improve the chances of visa certification
- **Region of employment (such as Midwest and Europe**) has a moderate influence on visa approvals. Different regions may have varying demand or labor shortages that influence decisions. Employers in regions with higher labor demand (such as the Midwest) may benefit from better odds of visa certification. Companies should be aware of regional trends when planning recruitment from abroad and collaborate with OFLC to understand how geographic preferences impact approval rates.
- **Full-time positions** have a slight impact on the approval process, though it's not a dominant factor. While full-time positions are generally more favorable, companies should still prioritize experience, education, and prevailing wages over full-time status when making visa applications.

- Implement **Gradient Boosting (Undersampled data) model** to predict the likelihood of visa certification based on applicant attributes. Such a model can assist OFLC in automatically shortlisting high-potential candidates, reducing the manual review workload.

Conclusion:

The analysis identifies key factors influencing visa certification, providing actionable insights that can guide the development of more effective visa processing strategies. By focusing on education, experience, training requirements, and wage adjustments, EasyVisa can help streamline the visa approval process, ensuring that the U.S. attracts the right talent to meet its workforce needs. Future steps include developing predictive models and optimizing application review workflows to further enhance the visa certification process.

These insights will help companies stay competitive by attracting the best talent from abroad while navigating the growing complexity of the visa approval process efficiently.